

MASTER SERIES I MICRO-METER “AIRLESS” LIQUID APPLICATOR[®]

Sprays with metered accuracy oils,
anti-oxidants, vitamins, flavors, tack
and release agents, etc.

ORIGINAL INSTRUCTIONS

SPRAY DYNAMICS

SERIES I MICRO-METER “AIRLESS” LIQUID APPLICATOR®

CONFIGURATION

This sheet defines the Master Series system for which this manual is supplied. In the event of future needs for parts or service, information on this sheet will assist our Service Technicians and help you get the unit back on-line. Do not remove this sheet from this manual except to make a copy for your service files; then be sure to return this original to the manual.

Master Series I® Liquid Applicator

SERIAL NO.: 1388

SALES ORDER: 2020053566

MODEL NAME: MSI APPLICATOR

PUMP TYPE: MODEL #44

NOZZLE TYPE: N250-60

POWER
REQUIREMENTS: 480 VAC / 3 PH / 60 HZ

CONTROL PANEL: 32095129 SERIAL #1432

SUPPLY VESSEL: 40 QUART STOCKPOT

IMPORTANT INFORMATION FOR ORDERING PARTS:

When ordering parts, be sure to list name of part,
part number, quantity required, and Machine Serial Number.

SPRAY DYNAMICS

SERIES I MICRO-METER “AIRLESS” LIQUID APPLICATOR®

PREFACE

SAFETY MEASURES

Throughout this manual there are 2 symbols next to information which should be read and carefully understood before doing any maintenance or operating the machine. The symbols are



Electrical warning



General warning

All operators should be adequately trained and specialized personnel should carry out any repairs beyond the regular service of the machine. These would include maintenance personnel, and line managers.



Before opening any electrical panel or junction box, proper Lock-Out/Tag-Out procedures should be followed. This includes pneumatics. Only competent persons should maintain electrical equipment.



Ear plugs and safety glasses should be worn when in the vicinity of this machine.



The airborne noise levels for the liquid applicator are dependent upon 3 variables. The number of pumps being used; the amount of air pressure used to get the required spray pattern; and the liquid being applied. Most liquid applicators will be in the range of 65-90 decibels.



Use safe lifting practices when installing equipment. Do not stand under equipment being lifted.

SPRAY DYNAMICS

SERIES I MICRO-METER “AIRLESS” LIQUID APPLICATOR®



System uses compressed air - check and maintain all connections every 6 months to prevent danger from burst / disconnected lines.



Ensure any oil on floor is cleaned up to prevent slip hazard.



Press system stop button and close pneumatic shut off valve in case of hose failure.



Set machine to proper pulsation rate and the correct air pressure given in manual to minimize irritation due to oil misting. Only operate applicator when correctly positioned in coating drum or hood.

SPRAY DYNAMICS

SERIES I MICRO-METER “AIRLESS” LIQUID APPLICATOR®

EQUIPMENT DESCRIPTION

PRINCIPLE OF OPERATION

This equipment is designed to apply a uniform coating of oil or other liquid materials to a band, belt or product traveling past the spray nozzles within a coating drum. The amount of product applied is accurately measured by positive displacement pneumatically operated pumps. The volume of the pumps as well as the number of strokes per minute is adjustable. The amount of liquid applied is adjustable over a wide range with this system. Very light coatings can be applied because the machine generates a series of overlapping round patterns rather than a constant spray. The number of pumps and nozzles varies with the size of the system. The control system to regulate the actuation of the pumps is a P.L.C. This, combined with the adjustable micrometer knob on the pump give the spray unit an unprecedented degree of flexibility.

This equipment is not intended to apply any type of slurry which contains solid particulates, flammable liquid, other hazardous liquids, or any liquids outside the scope of the original contract.

Any variations in the design of this system will void any implied warranty, unless authorized by Spray Dynamics. Any new products to be sprayed must be assessed for the potential risks and hazards and suitable measures must be taken.



Be aware that this machine sprays oil using air pressure. DO NOT remove the black flexible product line from the spray arm while the machine is running. Eye injury could occur if the oil were sprayed into the eyes. Wear safety glasses.

CONSTRUCTION

This equipment is constructed of 304 stainless steel, using sanitary design criteria. All components are designed and mounted for ease of maintenance and maximum flexibility. Stainless steel adjustable mounting brackets are supplied. This complete unit mounts to a 1.5" schedule 40 pipe.

SPRAY DYNAMICS

SERIES I MICRO-METER "AIRLESS" LIQUID APPLICATOR®

COMPONENTS

This system consists of four basic pieces of equipment: The pump manifold assembly, the spray arm assembly, and the control box.

Maintenance required would be before each shift - checking oil level in lubricator bowl and draining the filter bowl of excess liquid. Also, check that air pressures are set to proper levels.

The pump manifold assembly is a length of pipe to which the pumps are attached. The pipe acts as a supply reservoir for the pumps. The manifold must be full in order to maintain a consistent flow to the pump. It should not, however, be pressurized to more than 5 P.S.I. as liquid could be forced past the pump "O" Rings or check valve. The pumps should always be mounted below the manifold to prevent any possibility of the pumps sucking air.

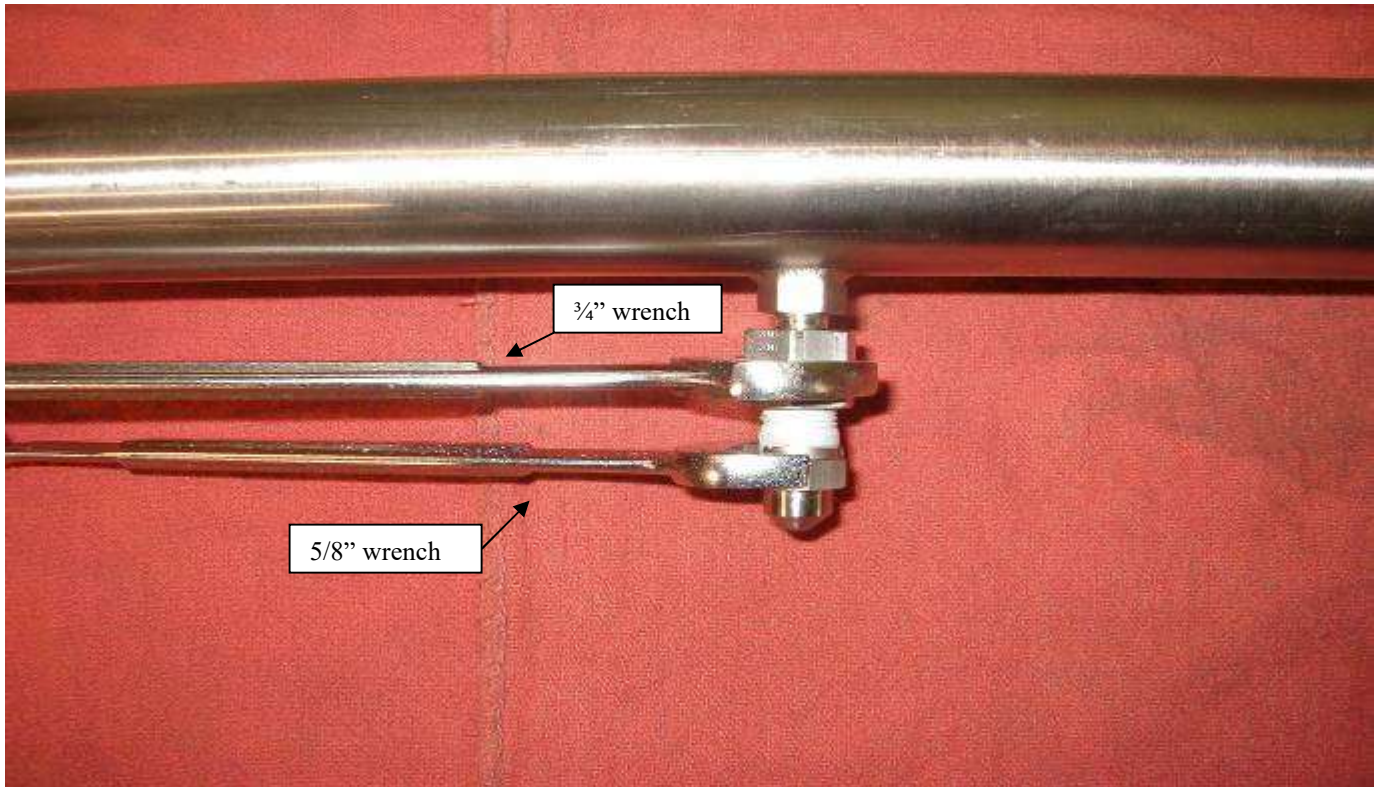
The Model 40 Series of pumps is engineered to handle various ranges of flow rates and all have the same basic construction. The pump must not be mounted with the adjustment assembly above the horizontal position. This would allow air to be entrapped within the pump.



The only maintenance normally required is the replacement of "O" Rings. This should be done every 2,500 hours when the fluid being pumped is oil. Liquids other than oil will require different maintenance schedules. The most critical seal is the piston rod "O" Ring; it should be replaced if air leaks into the product. This will cause a hollow or erratic nozzle spray pattern. The air piston "O" Ring is also a dynamic seal and will show wear. If it leaks, it can be detected leaking from the exhaust filter. This is a "floating" type "O" Ring. It may appear over-size with a loose fit on the piston; this is normal. It is designed for a low breakout friction and long life. All "O" Rings should be coated with oil or grease for installation.

The spray arm assembly consists of the length of pipe that extends into the coating drum and actually sprays the liquid. The arm is internally plumbed with 1/4" stainless steel tubing. This tubing runs from a fitting at one end to the nozzle tip at the other.

SPRAY DYNAMICS
SERIES I MICRO-METER “AIRLESS” LIQUID APPLICATOR®
LIQUID APPLICATOR SPRAY ARM



When installing or removing nozzles, use a $\frac{3}{4}$ " wrench to hold the spray arm nut from turning when you tighten the nozzle. Failure to hold the spray arm nut from turning will pinch the inner $\frac{1}{4}$ " tubing causing premature failure in the product line, voiding any warranty.

SPRAY DYNAMICS

SERIES I MICRO-METER “AIRLESS” LIQUID APPLICATOR®

LIQUID APPLICATOR SPRAY ARM



Close up of spray arm nozzle assembly. The gap between the two nuts is important. If they are not tightened properly, they will leak.



The only other maintenance required is a periodic check of the nozzle tip for a consistent spray pattern. The nozzle tip is a mechanical moving part and in time it may be necessary to replace it. (Refer to parts list).

When ordering parts please use the Spray Dynamics part number whenever possible to assure a fast and accurate turn-around.

Connecting the two assemblies is a cantilever arm arrangement which allows for the maximum flexibility in set-up. Liquid connection is by braided stainless steel hoses (one per pump).

SPRAY DYNAMICS

SERIES I MICRO-METER "AIRLESS" LIQUID APPLICATOR®

INSTALLATION

REFER TO INSTALLATION DRAWING



It is recommended that 2 people install the components of this machine at its final location. Enough space must be given to allow for installation and removal of spray arm from coating drum. The spray manifold should have enough room on each side of the manifold for dismantling and cleaning.

1. Attach a 1.5" schedule 40 post (minimum height 40") to in-feed end of tumble drum frame. Per Assembly Drawings.
2. Mount Control Box to post. Attach the FRL (filter-regulator-lubricator) assembly below the control panel. The FRL controls the pump actuations using air pressure.
3. Mount pump manifold and supply reservoir as shown in General Layout Drawing.
4. Mount cantilever bar to cross bar as shown. Position spray bar as shown such that nozzles spray onto product bed - when drum is running. Do not allow spray to hit walls of drum.
5. Connect black flex lines from check valve (on the bottom of the pump) to the spray arm.

Connect impolene tubing from FRL solenoid assembly to air manifold (Refer to drawing).



Connect main power to Control Box. Refer to the electrical drawing for your specific voltage. Once voltage has been applied to the control panel, check for continuity to ensure proper bonding and grounding of the circuitry. Check that the voltage is correct for this control panel.

Connect oil supply to pump manifold.

***** SYSTEM IS NOW READY TO OPERATE *****

SPRAY DYNAMICS

SERIES I MICRO-METER “AIRLESS” LIQUID APPLICATOR®

OPERATION

INITIAL STARTING PROCEDURE

1. Fill the supply tank or stockpot. (Be sure pump manifold is full of product, though not under more than 5 pounds of pressure).
2. Fill the air lubricator (PNEU LUBE).
3. Turn all pump adjusting knobs to minimum full clockwise position.
4. Open each purge nut located on the pumps, allowing entrapped air to escape. Close purge nut when oil spurts out.
5. Set the pressure regulator on the FRL. 80 P.S.I. is the recommended **starting pressure**.

NOTE: The above P.S.I. requirements vary from product to product.

6. For units supplied with a **mixer** in the stockpot, ensure the safety switch is engaged and press the **Reset** button. If the lid is off the stockpot, then the safety circuit will not allow the system to be restarted.
7. Adjust pulses per minute on panel interface to cycle the pumps at the required rate and adjust micrometer to proper setting and turn on unit. Allow unit to run until proper spray pattern is achieved.
8. Check spray pattern to be sure you have even coverage over the entire spray area.
9. Adjust pressure regulator up or down until even coverage has been achieved.
10. Lock pressure regulator at adjusted pressure, then document and label for future reference.



If the oil/liquid being applied causes excessive misting into the atmosphere, first try to reduce the air pressure, if this does not work in reducing the mist/overspray then some type of mist hood may be installed above application zone or at the infeed and discharge of the coating drum.

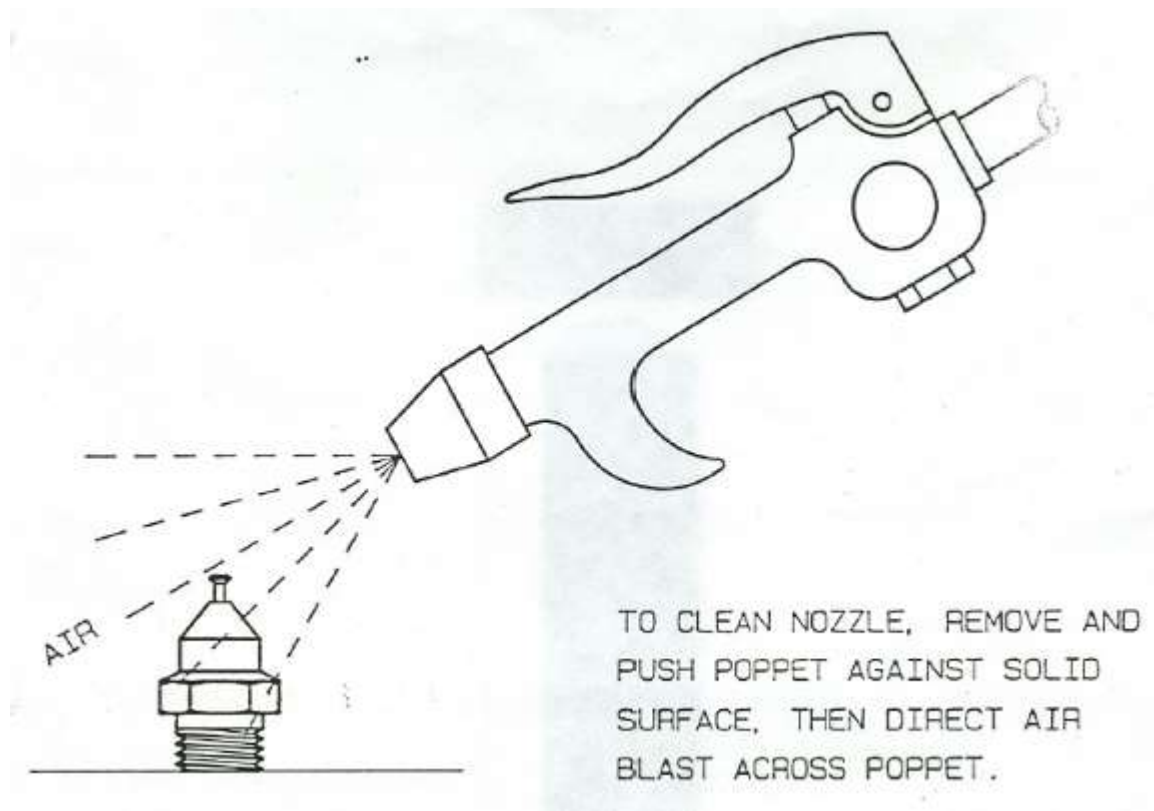
SPRAY DYNAMICS

SERIES I MICRO-METER "AIRLESS" LIQUID APPLICATOR®

OPERATION (Continued)

REGULAR OPERATING PROCEDURE

1. Check the air system daily. Drain the air filters and refill the lubricator. Pump volume and speed adjustments are not normally changed day to day. Check the supply tank and fill if required.
2. Shut machine down by turning off main air supply and power on machines with electronic controls. For long shutdown periods, drain the product from the machine, flush system by pumping mineral oil through each pump and nozzle.
3. If a nozzle drips, or does not spray properly, the cause usually is entrapped air or foreign particles, under the nozzle poppet seat. First, bleed any entrapped air out of the pump through the purge fitting. If this does not cure the problem, remove the nozzle and clean out as shown.



SPRAY DYNAMICS

SERIES I MICRO-METER “AIRLESS” LIQUID APPLICATOR®

REGULAR CLEANING

1. Drain Product from System
2. Fill stock pot or tank with hot soapy water and pump through the system to clean.
3. Fill with Hot Water and pump through the system to rinse.
4. Drain Water from all lines.
5. Blow Excess Water OFF with air nozzle

NOTE: Control boxes are wash down NOT high-pressure hose down.

DO NOT use High Pressure Hose on Control Boxes or solenoid valves.

SANITATION

Machinery should be cleaned and sanitized on a regular schedule to prevent any microbial formations. Each facility should have its own schedule for cleaning and products used for this purpose. ZEP superior solutions has several products used for cleaning, degreasing, and sanitizing equipment in the food industry with offices in France, Italy, Holland, and the United Kingdom.

SPRAY DYNAMICS

SERIES I MICRO-METER “AIRLESS” LIQUID APPLICATOR®

MAINTENANCE



Before opening any electrical panel or junction box, proper Lock-Out/Tag-Out procedures should be followed. This includes pneumatics. Only competent persons should maintain electrical equipment.



System uses compressed air - check and maintain all connections every 6 months to prevent danger from burst / disconnected lines.



Inspect oil pipework and connection on a weekly basis. Repair as required.



Inspect oil hoses and connections on a monthly basis. Repair or replace as required.



Set machine to proper pulsation rate and the correct air pressure given in manual to minimize irritation due to oil misting.

SPRAY DYNAMICS

SERIES I MICRO-METER “AIRLESS” LIQUID APPLICATOR®

MAINTENANCE (Continued)

MINOR:

DAILY: Check Air line Lubrication

Check Air line Pressure (40 - 80 PSI as determined at initial startup)



MAJOR: 2500 Hours: Rebuild Pumps

(Note: 2500 Hours: When pumping oil; water and other product as needed)

1. Turn off air
2. Drain manifold
3. Remove pumps from manifold
4. Check for play between piston and bushing, if excessive play is present send pumps to Spray Dynamics for rebuild. Call for price and scheduling.
5. Replace o-rings (Use food grade grease to lubricate o-rings.)

SPRAY DYNAMICS

SERIES I MICRO-METER "AIRLESS" LIQUID APPLICATOR®

REPLACING THE PRIMARY O-RING

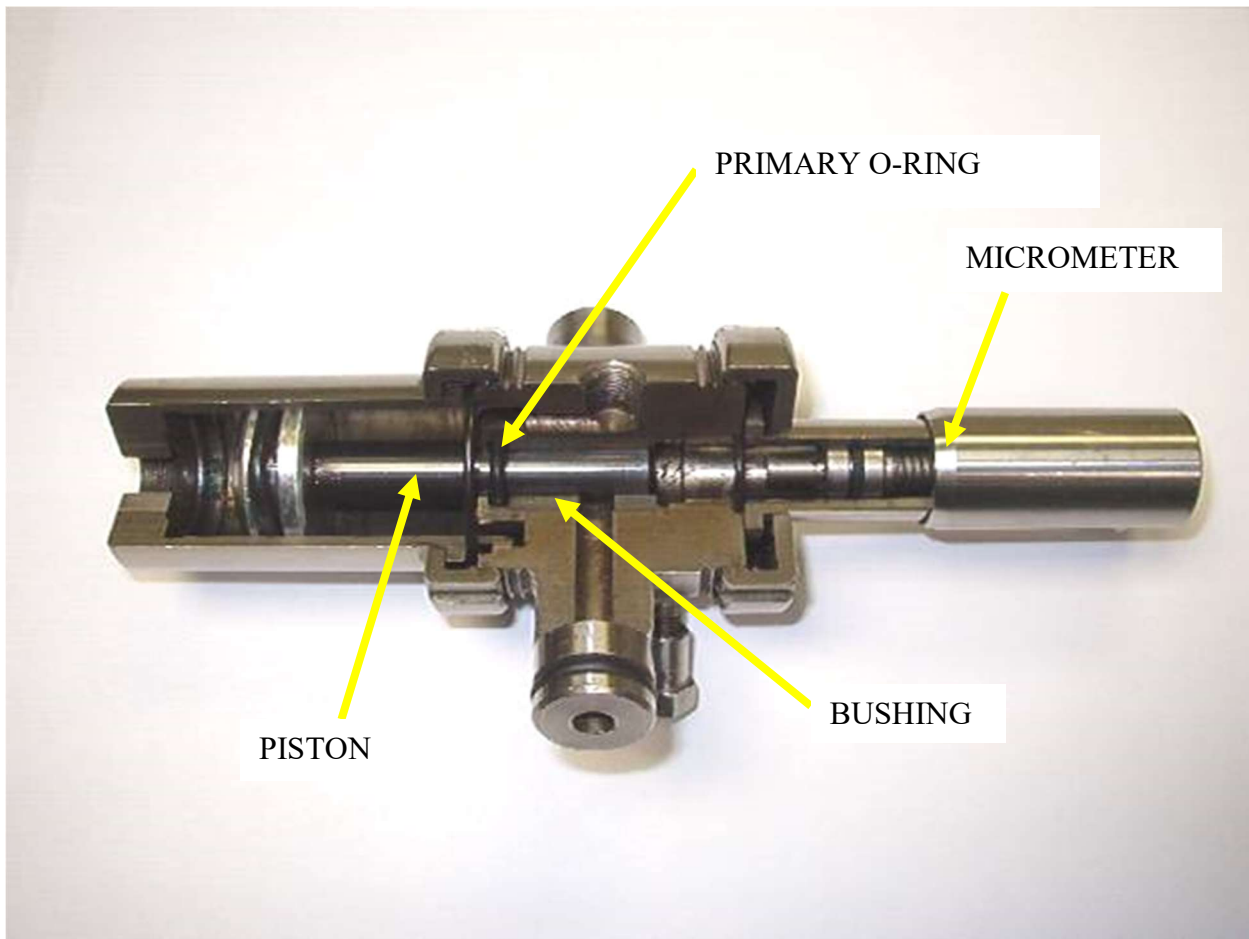


FIGURE 1

Figure 1 shows a cutaway view of the internals of a typical MSI pump. The Primary o-ring is seated in a groove in the bushing that guides the piston. To replace the o-ring first disassemble the pump and remove the o-ring using an o-ring removal tool, a small pick or a small screwdriver.

SPRAY DYNAMICS

SERIES I MICRO-METER "AIRLESS" LIQUID APPLICATOR®

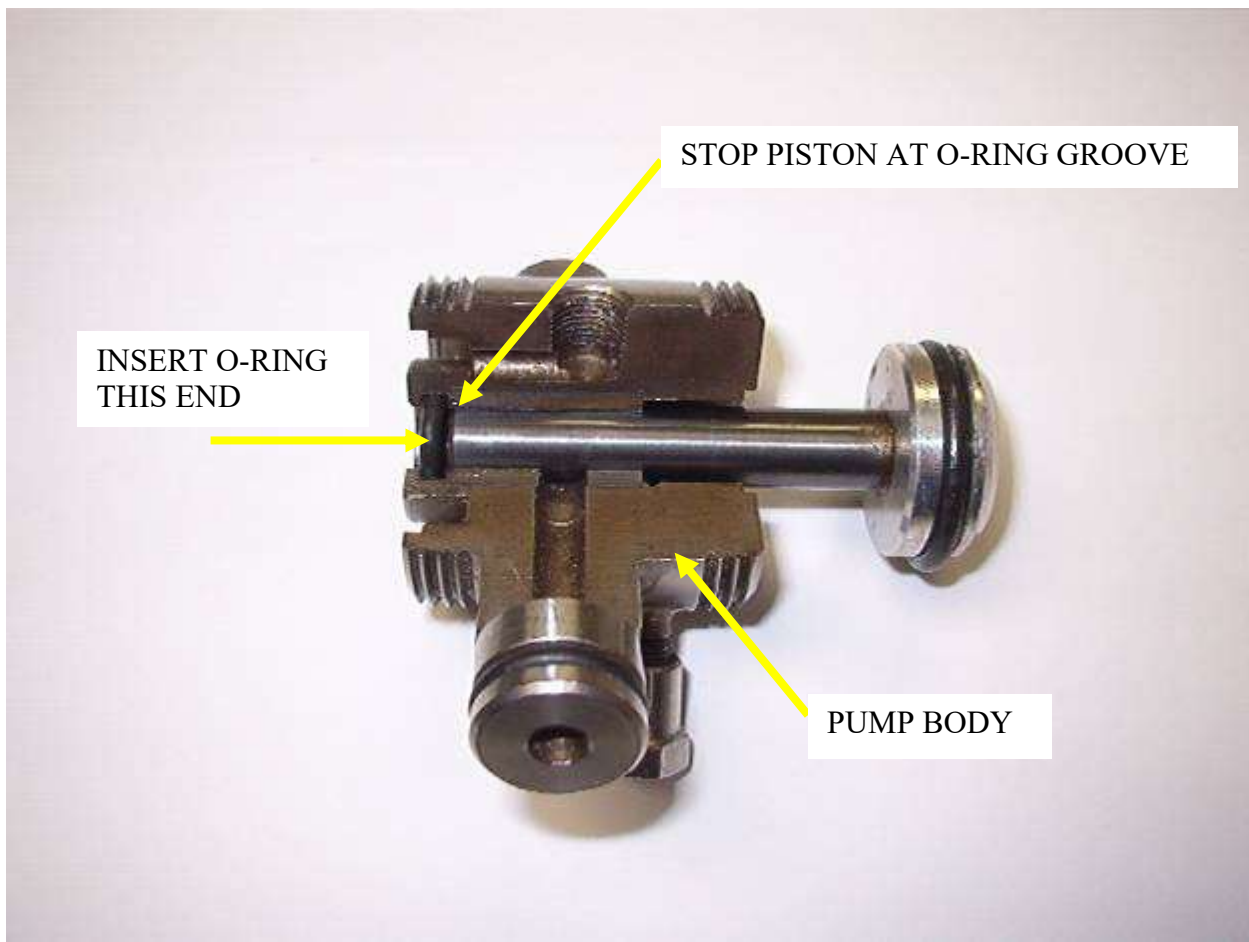


FIGURE 2

When replacing the o-ring first insert the piston from the opposite end of the bushing and stop it just as it gets to the o-ring groove as shown in figure 2. Holding the body and piston in one hand you can then push the o-ring in with the other hand using a blunt tool. The piston creates a back stop to help position the o-ring into the groove. Be careful not to nick or cut the o-ring. If you do the o-ring will fail immediately. Once the o-ring is in place remove the piston and re-assemble the pump. Before re-inserting the piston first lubricate the end of the piston. Then push and gently twist the piston into position being careful not to nick or pinch the o-ring.

SPRAY DYNAMICS

SERIES I MICRO-METER "AIRLESS" LIQUID APPLICATOR®

TROUBLESHOOTING

	<u>Symptom</u>	<u>Check</u>
Power ON	No Spray	Micrometer setting on zero Air Pressure Check Fuse Air Lock in Pump Manifold
	Poor Spray Pattern	Air Pressure Clean Nozzles
	Erratic Spray	Air Lines may be too long Check Valve O-ring
	Solenoid Click/No Spray	Turn On Air
Power On / Air ON	System on Standby Air from Muffler	Large Piston O-Ring
	System on Standby Air in Product	Small Piston O-Ring

LONG TERM STORAGE (OVER SIX MONTHS)

LIQUID APPLICATOR Flush mineral oil through system (including nozzles).
Before use replace "O" Rings and flush with hot water.

SPRAY DYNAMICS
SERIES I MICRO-METER "AIRLESS" LIQUID APPLICATOR®
LIQUID APPLICATOR SYSTEM - 44 PUMP
RECOMMENDED SPARE PARTS LIST
FOR A 3 PUMP SYSTEM

<u>PART #</u>	<u>QTY</u>	<u>DESCRIPTION</u>
20605083	1*	Coil, Soleniod,24VDC,FEST#8030802
20616591	1*	Valve, Solenoid,G3/8,VUVS,FEST#575594
20562879	3*	Viton O'Ring for check valve
20547090	3	Check Valve Spring
20532434	3*	Nozzle (N250-60 Degree)
20503076	1*	Pump Wrench
20547096	1	#44 Pump Complete
20503073	3	#44 Piston Assembly
20503078	3*	#44 Viton O'Ring Kit
31133010	2*	PNEU-LUBE Food Grade Lubricating Oil (16 ounce bottles - per case of 6 bottles)

***HIGHLY RECOMMENDED TO HAVE ON HAND**

REV 04/21